# Sai Charan Reddy

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## Summary

Computer Science student specializing in Artificial Intelligence, combining software engineering with AI application development. Experienced in Java, Spring Boot, React, Python, FastAPI, LLMs, RAG, hybrid retrieval, multi-agent workflows, REST APIs, databases, and cloud deployment. Focused on building scalable, reliable applications through strong problem-solving and software engineering fundamentals.

## Education

**Bennett University, Greater Noida**
*B.Tech. in Computer Science* — *Sep 2023 – Present*

## Technical Skills

**Languages**: Python, Java, JavaScript
**Software Development**: Spring Boot, Spring Security, React, FastAPI, REST APIs, JPA/Hibernate, JWT Authentication, Docker
**AI/ML**: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Prompt Engineering, Transformers, Embeddings, Hybrid Search, Multi-Agent Systems
**Frameworks & Libraries**: PyTorch, LangChain, LangGraph, NumPy, Pandas, Gemini, BAAI/bge, SPLADE++
**Data, Cloud & Tools**: PostgreSQL, MongoDB, Qdrant, AWS (EC2, S3), Git/GitHub, LangSmith
**Core Concepts**: Data Structures & Algorithms, Object-Oriented Programming

## Projects

**Hirely – Job Platform** — **GitHub** | **Live Demo**
*Java, Spring Boot, React, PostgreSQL, AWS* — *2026 – Present*

- Built a full-stack job platform for Candidate, Company, and Hiring Manager roles using layered Spring Boot architecture, PostgreSQL/JPA, and MapStruct.
- Implemented JWT authentication, BCrypt password hashing, role-based authorization, asynchronous email notifications, and resume MIME validation; deployed the backend on AWS EC2 and frontend on Vercel.

**Task Tracker** — **GitHub**
*Java, Spring Boot, React, PostgreSQL* — *Apr 2026*

- Developed a full-stack task manager with a Spring Boot 3 REST API, React/Vite frontend, and PostgreSQL persistence.
- Implemented Spring Security authentication, user-task JPA relationships, and REST endpoints for CRUD, per-user retrieval, deadlines, and completion status.

**SkillLens – AI Resume Screening** — **GitHub**
*Python, FastAPI, Gemini, Qdrant, Hybrid Retrieval* — *2026 – Present*

- Built a resume-screening pipeline combining dense BAAI/bge-base-en-v1.5 and sparse SPLADE++ embeddings for hybrid retrieval beyond keyword matching.
- Ranked resumes using top-three chunk similarity scores and integrated Gemini for query enhancement and automatic inference of requested result counts.

**Float Chat – Multi-Agent Data Retrieval** — **GitHub**
*Python, LangGraph, Gemini, MongoDB, PostgreSQL* — *2025*

- Architected a three-agent LangGraph workflow (Decomposition, Filter, SQL) to route natural-language queries to appropriate retrieval operations.
- Combined MongoDB geospatial/metadata filtering with PostgreSQL structured oceanographic float-profile queries, matching retrieval methods to the data type.

**GPT-2 Implementation from Scratch** — **GitHub**
*Python, PyTorch, Transformers* — *2025*

- Implemented GPT-2 in PyTorch with token/positional embeddings, causal multi-head self-attention, pre-norm transformer blocks, and residual connections.
- Trained on FineWeb using cosine learning-rate decay, gradient accumulation, and bfloat16 mixed precision; added KV-cache prefill/decode and achieved 2.77 validation loss.

## Certifications

Foundations of Deep Learning: Concepts and Applications | Data Structures | AI Workflow: Data Analysis and Hypothesis Testing